

F. The Fuel

Notes

- A fuel is any material that can be made to react with other substances so that it releases energy as heat energy or to be used for work. Some common fuels are wood, coal, petrol, diesel etc. Uranium and Plutonium are nuclear fuels used to produce nuclear energy.

Ideal fuel :

- An ideal fuel is a fuel which produces adequate amount of heat and very less amount of pollutants.
- An ideal fuel has following characteristics—
 - It is readily available.
 - It is less expensive.
 - It burns easily in air at a moderate rate.
 - It produces a large amount of heat.
 - It can transport easily.
 - It has high calorific value.
 - It does not leave behind any undesirable substances.

Heating value of fuel :

- Heating value or calorific value of fuel is the amount of heat produced by a complete combustion of fuel and it is measured as a unit of energy per unit mass or volume of substance (e.g. K cal/kg, KJ/kg, J/mol and Btu/m³).

Heating value of some fuels

Fuel	Heating value (KJ/kg)
Cow dung cake	6,000 – 8,000
Wood	17,000 – 22,000
Coal	25,000 – 33,000
Petrol	45,000
Kerosine oil	45,000
Diesel	45,000
Methane	50,000
CNG	50,000
LPG	55,000
Biogas	35,000 – 40,000
Hydrogen	1,50,000

- The heat energy released by reactions of fuels is converted into mechanical energy via an engine. Other times the heat itself is valued for warmth, cooking, illumination, in industrial processes etc.

Harmful effects of burning of fuel :

- The burning of fuels like wood, coal and petroleum products releases unburned carbon particles in the air. These fine carbon particles are dangerous pollutants which can cause respiratory diseases.
 - Incomplete combustion of fuels produces a very poisonous gas called carbon monoxide. Excessive inhaling of carbon monoxide gas can kill a person. We should never sleep in a room with closed door and windows and having a coal fire burning inside. This is because when coal burns in an insufficient supply of air in the room, then a lot of carbon monoxide is produced and when the persons sleeping in that room breathe in poisonous carbon monoxide gas, they may all die.
 - Burning of fuels releases carbon dioxide in the environment. Increased percentage of carbon dioxide in air is causing global warming. Carbon dioxide gas in the air traps sun's heat by producing greenhouse effect.
 - Burning of coal, petrol and diesel produces sulphur dioxide which goes into the air. It is an extremely suffocating and corrosive gas. It may damage our lungs. The burning of petrol and diesel in the engines of vehicles also releases oxides of nitrogen in the air. These oxides produced by the burning of fuels dissolve in rain water and form acids. The rain water containing acid is called acid rain, which is very harmful to crops, buildings and soil.
- The use of petrol and diesel as fuels in automobiles is being replaced by **compressed natural gas (CNG)** because—
 - It produces very small amount of harmful gas.
 - It burns without smoke, so it is a clean fuel.
 - Disadvantages of burning wood as fuel are as follows :
 - It produces a lot of smoke which is very harmful for human beings.
 - When trees are cut down to obtain wood for use as fuel, then all the useful substances which are obtained from trees are lost.
 - Deforestation is very harmful for environment.
 - LPG** (Liquefied petroleum gas) is a better domestic fuel than wood due to :
 - It has higher calorific value, so it produces much more heat on burning than an equal mass of wood.
 - It burns completely without leaving behind any solid residue but wood leaves behind a lot of ash on burning.

Fossil fuel :

- Fossil fuel is a general term for buried combustible geologic deposits of organic materials, formed from decayed plants and animals that have been converted to crude oil, coal, natural gas, or heavy oils by exposure to heat and pressure in the earth's crust over hundreds of millions of years.
- Fossil fuel are non-renewable sources of energy. As we are excessively dependent on these, alternative forms of energy need to be developed in order to deal with the imminent energy scarcity. These alternative sources of energy also need to be more eco-friendly.
- The main fossil fuels are as follows—

(a) Coal :

- Coal is one of the most important fossil fuel.
- Coal is solid carbon rich material. In addition to carbon it contains hydrogen, sulphur, oxygen & nitrogen.
- Coal is usually brown or black.
- Coal often occurs in stratified sedimentary deposits.
- On burning in air, coal produces mainly carbon dioxide gas.
- By processing of coal some beneficial products are obtained as coke, coal tar and coal gas.
- There are 4 main types of coal –
 1. Peat coal,
 2. Lignite coal (brown coal),
 3. Bituminous coal (soft coal), and
 4. Anthracite coal (hard coal)
- **Peat coal** has excessive quantity of moisture and impurities. Peat coal is the lowest grade of coal.
- **Lignite** is a soft, brown, combustible, sedimentary rocks formed from naturally compressed peat.
- **Bituminous coal** or black coal is a relatively soft coal containing a tar like substance **bitumen** or **asphalt**.
- It is of higher quality than lignite coal, but of poorer quality than anthracite.
- It is formed usually as the result of high pressure being exerted on lignite.
- **Anthracite coal** often referred to as hard coal, is a hard, compact variety of coal that has a submetallic lustre.
- It has the highest carbon content, the fewest impurities.
- It is the highest ranking of coals.
- It is with low sulphur content, that makes it an good burning fuel.

(b) Coke :

- Coke is a fuel with few impurities and a high carbon content.

- It is a solid, hard, porous and black material derived from destructive distillation of low ash, low sulphur bituminous coal in the absence of oxygen.
- It is used in industrial manufacturing of steel and in extraction of several metals.

(c) Coal tar :

- Coal tar is a thick dark liquid with bad smell.
- It is a by product of the production of coke and coal gas from coal.
- It has both medical and industrial uses.
 1. Medical uses of coal tar :
 - It may be applied to the affected area to treat psoriasis and seborrheic dermatitis.
 2. Industrial uses of coal tar :
 - It is used in preparation of synthetic colors, explosive substances, plastics, photographic material and road manufacturing materials etc.

(d) Coal gas :

- Coal gas is a flammable gaseous fuel made from coal.
- It is produced when coal is heated strongly in the absence of air.
- The main components of coal gas are hydrogen, methane and carbon monoxide.
- It is used as fuel in different industries established near to the coal process plants.

(e) Petroleum :

- Petroleum is an oily, thick, flammable, usually dark coloured liquid that is a form of bitumen or a mixture of various hydrocarbons, naturally occurring in different parts of world, beneath the earth surface.
- It is also known as black gold.
- It also contains sulphur, oxygen and nitrogen in small quantity.
- It is also known as traditional oil or light oil.
- The word petroleum comes from the Latin *petra* meaning rock and *oleum* meaning oil.
- It can be separated into fractions including natural gas, gasoline, naphtha, kerosene, fuel and lubricating oils, paraffin wax and asphalt.
- **Petrochemicals** are chemical products derived from petroleum as acetylene, benzene, ethane etc., from which other chemicals, pesticides, plastics, synthetic fibres, paint and medicines are derived.

(f) Natural Gas :

- Natural gas is a fossil fuel.
- Like other fossil fuels such as coal and oil, natural gas formed from the buried plants, animals and micro-organisms that lived millions of years ago.

- Natural gas is a mixture of gaseous hydrocarbons as methane, ethane, propane and butane and other gases like nitrogen, carbon dioxide and hydrogen sulphide.
- Natural gas is found in deep underground rocks formations or associated with other hydrocarbon reservoirs.
- **Helium** is also found in natural gas of some area.
- The percentage of methane in natural gas is highest i.e. about 45-85%.
- Natural gas used as a source of energy for heating, cooking, and electricity generation. It is also used as fuel for vehicles and as a chemical feed stock in the manufacturing of plastics and other commercially important organic chemicals.

(g) Liquefied Petroleum Gas (LPG) :

- LPG is made during natural gas processing and oil refining.
- Propane and butane are separated from the natural gas and liquefied at high pressure and stored in cylinders or tanks.
- It is used as fuel in houses and hotels.
- LPG gas is odourless. To detect its leakage, **ethyl mercaptan** (a clear liquid with a distinct odour) is mixed.

(h) Compressed Natural Gas (CNG) :

- In CNG mainly methane is stored at high pressure.
- It is a fuel which can be used in place of gasoline, diesel fuel and LPG.
- CNG combustion produces fewer undesirable gases than the fuels mentioned above. So it is eco-friendly fuel or ideal fuel.

Question Bank

1. Which of the following is a natural fuel?

- (a) Coal gas (b) Tar
(c) Coke (d) Petroleum

U.P.P.C.S. (Pre) 1990

Ans. (d)

The petroleum products are subject to natural fuel. They are made from the fossils buried within the Earth. Other options coke and tar can be achieved by destructive distillation of coal or wood, while coal gas can be created artificially.

2. Which of the following fossil fuels is the cleanest fuel ?

- (a) Coal (b) Petrol
(c) Natural gas (d) Diesel

U.P.P.S.C. (R.I.) 2014

Ans. (c)

The use of natural gas in power generation provides a cleaner alternative to coal and other fossil fuels, reducing carbon and other emissions and resulting in both immediate and long-term benefits for public health and the environment.

3. Which of the following is a fossil fuel?

- (a) Alcohol (b) Ether
(c) Water gas (d) Natural gas

U.P.P.C.S. (Mains) 2016

Ans. (d)

Fossil fuels, including coal, oil and natural gas are currently the world's primary energy sources. The fossil fuels formed from the organic material over the course of million of years have fuelled the global economy over the past century. Yet the fossil fuels are finite resources and they can also irreparably harm the environment.

4. Which one of the following has the highest fuel value ?

- (a) Hydrogen (b) Charcoal
(c) Natural gas (d) Gasoline

I.A.S. (Pre) 1997

Ans. (a)

The heating value or calorific value of a fuel is the amount of heat that is released after a specific amount of fuel is burnt completely in air or oxygen. It is generally expressed in kilojoule per kg (KJ/kg) or megajoule per kg (MJ/kg). Hydrogen has the highest heating value (about 150,000 KJ/kg) among all fuels. Hydrogen is used as a rocket fuel and in burner producing high temperature.

5. With reference to coal-based thermal power plants in India, consider the following statements :

1. None of them uses seawater.
2. None of them is set up in water-stressed district.
3. None of them is privately owned.

How many of the above statements are correct?

- (a) Only one (b) Only two
(c) All three (d) None

I.A.S. (Pre) 2023

Ans. (d)

Many coastal power plants in India use seawater for cooling purposes. For instance, the Mundra Thermal Power Station in Gujarat, India, which is one of the largest coal-based power plants in the country, uses seawater for cooling. As per the Centre for Science and Environment (CSE) Study of 2021, about 48% of existing coal-based power plants in India are located in water-scarce districts. There are both privately and publicly owned coal-fired thermal power stations in India. Many private corporations have invested in the power industry and run coal-fired thermal power facilities, such as Adani Power Limited which is the largest private thermal power producer in India. Hence, all three statements are incorrect.

6. Consider the following statements :

1. Coal ash contains arsenic, lead and mercury.
2. Coal-fired power plants release sulphur dioxide and oxides of nitrogen into the environment.
3. High ash content is observed in Indian coal.

Which of the statement given above is/are correct?

- (a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2020

Ans. (d)

Coal ash is the waste that is left after coal is combusted (burnt). It includes fly ash (fine powdery particles that are carried up the smoke stack and captured by pollution control devices) as well as coarser materials that fall to the bottom of the furnace. Most coal ash comes from coal-fired electric power plants. Depending on where coal was mined, coal ash typically contains heavy metals including arsenic, lead, mercury, cadmium, chromium and selenium, as well as other metals and elements. If eaten, drunk or inhaled, these toxicants can cause cancer and other serious health hazards. Hence, statement 1 is correct.

Depending on the source (type) of coal the emissions from the burned coal of coal-fired power plants contain pollutants such as sulphur dioxide (SO_2), sulphur trioxide (SO_3), oxides of nitrogen (NO_x), particulate matter (PM), mercury (Hg), trace metals, etc. Thus, statement 2 is correct.

Ash content of coal produced in India is generally 25 to 45% whereas ash content of imported coal varies from 10 to 20%. Indian coal has comparatively higher ash content than imported coal due to drift theory of formation of coal deposits in India. Therefore, statement 3 is also correct.

7. Why is there a concern about copper smelting plants?

1. They may release lethal quantities of carbon monoxide into environment.
2. The copper slag can cause the leaching of some heavy metals into environment.
3. They may release sulphur dioxide as a pollutant.

Select the correct answer using the code given below.

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2021

Ans. (b)

Copper smelting plants separate elemental copper from copper concentrates/ores through multiple sulphide oxidizing stages. The major air pollutants emitted from the process are sulphur dioxide and particulate matter (PM) and the main portion of the solid waste is discarded slag which typically contains heavy metals including arsenic, lead, cadmium, nickel etc. Thus, copper slag can cause leaching of some heavy metals into environment. Hence, statements 2 and 3 are correct.

Statement 1 is incorrect as copper smelting plants may release carbon monoxide and other air pollutant gases but they do not release lethal quantities of carbon monoxide into environment.

8. Which of the following causes maximum indoor chemical pollution?

- (a) Burning mosquito coil (b) Burning coal
(c) Burning cooking gas (d) Room spray

Jharkhand P.C.S. (Pre) 2021

Ans. (b)

Among the given options, burning of coal causes maximum indoor chemical pollution. Burning of coal produces many harmful chemical pollutants.

9. Consider the following statements :

Coke is one of the materials of the charge added to a blast furnace for the production of steel/iron. Its function is to

1. Act as the reducing agent
2. Remove silica associated with the iron ore
3. Function as fuel, to supply heat
4. Act as an oxidizing agent

Of these statements :

- (a) 1 and 2 are correct (b) 2 and 4 are correct
(c) 1 and 3 are correct (d) 3 and 4 are correct

I.A.S. (Pre) 1998

Ans. (c)

Coke is added to blast furnace for the production of iron/steel. It functions as fuel to supply heat. Besides supplying the heat, it also acts as a reducing agent by removing oxygen from iron ore (iron oxide) in the form of carbon monoxide and carbon dioxide.

10. With reference to furnace oil, consider the following statements :

1. It is a product of oil refineries.
2. Some industries use it to generate power.
3. Its use causes sulphur emissions into environment.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2021

Ans. (d)

Furnace oil (or fuel oil) is fuel consisting mainly of residues from crude oil distillation. Hence, statement 1 is correct.

Furnace oil is used as a fuel in different type of combustion equipments for power generation. It is used primarily for steam boilers in power plants, aboard ships, and in industrial plants. Hence, statement 2 is correct.

The oxides of sulphur (SO_x ; $\text{SO}_2 + \text{SO}_3$) emissions and secondary sulphate formation as particulate matter (PM) are a direct result of the sulphur content of the furnace oil. Hence, statement 3 is also correct.

11. Consider the following statements and choose the right options given below :

(1) Fuel obtained from plastic waste has low octane rating.

(2) Lower the octane number, more is the ability of the fuel to withstand compression without detonating.

- (a) Only the statement (1) is correct
- (b) Only the statement (2) is correct
- (c) Both the statements (1) and (2) are incorrect
- (d) Both the statements (1) and (2) are correct

Chhattisgarh P.C.S. (Pre) 2023

Ans. (c)

Fuel obtained from plastic waste has high octane rating. It contains no lead and is known as 'green fuel'. An octane rating, or octane number, is a standard measure of a fuel's ability to withstand compression in an internal combustion engine without undergoing pre-ignition. The higher the octane number, the more compression the fuel can withstand before detonating. Hence, both statements are incorrect.

12. Octane number is a measure of the quality of –

- (a) Edible oils
- (b) Petrol
- (c) Kerosene oil
- (d) Perfumed oil

U.P. Lower Sub. (Pre) 2009

Ans. (b)

The octane number of a fuel (petrol) is the measure of its antiknock quality when it is used in a spark ignition, internal combustion engine as compared to the antiknock quality of ISO-octane. The higher octane number of a fuel means the better anti-knocking capacity for that fuel.

13. The quality of petrol is expressed by :

- (a) Cetane number
- (b) Octane number
- (c) Butane number
- (d) Reynolds number
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (b)

See the explanation of above question.

14. Cetane number is used as a quality parameter mainly for which of the following?

- (a) Petrol
- (b) Kerosene
- (c) Diesel
- (d) Turpentine Oil

U.P. Lower Sub. (Pre) 2015

Ans. (c)

As the octane number is used to measure the ignition stability of gasoline (petrol), the cetane number is used as a quality parameter of diesel. The combustion of diesel without spark is provided by compression. Thus as soon as the diesel fuel is ignited and compressed the cetane number will be higher. This identifies the good quality of diesel.

15. The fuel used in a diesel engine :

- (a) Vapour of diesel and air
- (b) Diesel only
- (c) Mixture of diesel and petrol
- (d) Mixture of diesel, air and petrol

U.P.P.C.S. (Pre) 1994

Ans. (a)

A diesel engine is an internal combustion engine in which ignition in fuel is generated by using heat produced by compressing air in closed place. There is no carbureter in diesel engines for combustion as in petrol engines. In a diesel engine, the air inside a cylinder is compressed to such a level so that the temperature of air reaches around 80°F and this temperature works as combustion in diesel engines.

16. Cooking gas is mainly –

- (a) Carbon dioxide
- (b) Carbon monoxide
- (c) Methane
- (d) Nitrogen and Oxygen

44th B.P.S.C. (Pre) 2000

Ans. (c)

Methane is the chief component of biogas or gobar gas which is used for domestic purposes.

17. Which one of the following gases, released from biogas plant is used as a fuel gas?

- (a) Butane
- (b) Propane
- (c) Methane
- (d) Ethane

U.P. P.C.S. (Pre) 2016

Ans. (c)

Methane is the principal gas in biogas. It is also the main component of natural gas, a fossil fuel, and can be used to replace natural gas in many applications including cooking, heating, steam production, etc.

18. Biogas mainly consists of –

- (a) Carbon dioxide and Hydrogen
- (b) Hydrogen and Methane
- (c) Carbon dioxide and Methane
- (d) Hydrogen and Oxygen

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (c)

Biogas typically refers to a mixture of different gases produced by the breakdown of organic matter in the absence of oxygen. Biogas is primarily methane and carbon dioxide and may have small amounts of hydrogen sulfide, moisture and siloxanes. Biogas or gobar gas generally contains about 50%-75% Methane and about 25%-45% carbon dioxide.

19. Which among the following is the main constituent of biogas?

- (a) Methane
- (b) Propane

- (c) Butane (d) Ethane
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

See the explanation of above question.

20. The main component of biogas is :

- (a) Hydrogen (b) Methane
(c) Butane (d) Acetylene

Uttarakhand P.C.S. (Pre) 2012

Ans. (b)

See the explanation of above question.

21. The main component of Gobar Gas is –

- (a) Methane (b) Ethane
(c) Propane (d) Chlorine

U.P.P.S.C. (GIC) 2010

M.P.P.C.S. (Pre) 1999

Ans. (a)

See the explanation of above question.

22. Which of the following is the procedure for a biogas plant–

- (a) Fermentation (b) Reduction
(c) Hydrogenation (d) Polymerization

U.P. U.D.A./L.D.A. (Pre) 2013

Ans. (a)

The gohar gas formed from cattle dung and other wastes, has a lot of energy which can be stored for a biogas plant by the process of fermentation.

23. Who among the following invented 'Gobar Gas' system?

- (a) C.V. Raman (b) J.C. Bose
(c) C.B. Desai (d) H. Khorana

U.P.P.C.S. (Pre) 2015

Ans. (*)

The first pilot gas plant for producing methane from cattle dung and other cellulosic organic materials was first designed by S.V. Desai and set up at the Indian Agricultural Research Institute (IARI), New Delhi, during 1941-42.

24. The gas which emits from rice field is -

- (a) Ethane (b) Methane
(c) Nitrogen (d) All of the above

U.P.P.C.S. (Pre) 2006

Ans. (b)

Nitrous oxide, methane and carbon dioxide gases emit from rice field. Ethane, Methane and nitrogen are given in options hence option methane is the correct answer.

25. The main component of Liquid Petroleum Gas (LPG) are :

- (a) Methane, Ethane and Hexane
(b) Methane, Pentane and Hexane
(c) Ethane, Propane and Butane
(d) Methane, Carbon Monoxide and Hydrogen

U.P.P.C.S. (Mains) 2013

Ans. (c)

LPG is the abbreviation or short form for Liquefied Petroleum Gas. Like all fossil fuels, it is a non-renewable source of energy. It is extracted from crude oil and natural gas. The main composition of LPG are hydrocarbons containing three or four carbon atoms. The normal components of LPG thus are propane (C_3H_8) and butane (C_4H_{10}). LPG mixtures contain a little more propane in winter and a little more butane during summer but, on average, the makeup is approximately 65% butane and 35% propane. Small concentrations of other hydrocarbons like ethane may also be present. Ethyl mercaptan is normally used as a stenching agent for smell.

26. Which one of the following is the main constituent of LPG?

- (a) Hexane (b) Pentane
(c) Butane (d) Methane

U.P. P.C.S. (Pre) 2021

Ans. (c)

See the explanation of above question.

27. What is the main component of L.P.G. –

- (a) Methane (b) Ethane
(c) Propane (d) Butane

Chhattisgarh P.C.S. (Pre) 2005

Ans. (d)

See the explanation of above question.

28. LPG used as domestic fuel mainly contains –

- (a) Methane (b) Acetylene
(c) Ethylene (d) Butane

U.P.R.O./A.R.O. (Mains) 2013

Ans. (d)

See the explanation of above question.

29. LPG used as domestic fuel chiefly contains :

- (a) Methane (b) Propane
(c) Ethylene (d) Butane
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

67th B.P.S.C. (Pre) 2022

Ans. (e)

See the explanation of above question.

30. Indane gas is a mixture of –

- (a) Butane and Hydrogen (b) Butane and Oxygen
(c) Butane and Propane (d) Methane and Oxygen

U.P.P.C.S (Pre) 2011

Uttarakhand P.C.S. (Pre) 2010

R.A.S./R.T.S. (Pre) 1992

Ans. (c)

LPG is the flammable mixture of hydrocarbon gases used as fuel. Indane gas (LPG) is mainly the mixture of propane (C_3H_8) and butane (C_4H_{10}).

31. The stench due to leakage from LPG cylinder is because of :

- (a) Methane (b) Propane
(c) Butane (d) Ethyl mercaptan

U.P.P.C.S. (Pre) 2017

Ans. (d)

Liquefied Petroleum Gas (LPG) is a flammable mixture of hydrocarbon gases which is used as fuel. LPG is a mixture of butane and propane. LPG is an odourless gas product. However, to identify the leakage ethyl mercaptan is used in LPG.

32. Match items in the List-I with List- II and select the correct answer using the codes given below the lists :

List-I	List-II
(Fuel Gases)	(Major Constituents)
A. CNG	i. Carbon monoxide, Hydrogen
B. LPG	ii. Butane, Propane
C. Coal gas	iii. Methane, Ethane
D. Water gas	iv. Hydrogen, Methane, Carbon monoxide

Code :

	A	B	C	D
(a)	ii	i	iii	iv
(b)	ii	iv	iii	i
(c)	iii	ii	iv	i
(d)	iii	ii	i	iv

R.A.S./R.T.S.(Pre) 2013

Ans. (c)

The correctly matched lists are as follows –
CNG – Methane, Ethane
LPG – Butane, Propane
Coal gas – Hydrogen, Methane, Carbon monoxide
Water gas – Carbon monoxide, Hydrogen

33. Assertion (A) : The main component of the liquefied petroleum gas is methane.

Reason (R) : Methane can be used directly for burning in homes and factories where it can be supplied through pipelines.

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(c) (A) is true but (R) is false.

(d) (A) is false but (R) is true.

I.A.S. (Pre) 2005

Ans. (d)

The Assertion (A) of given question is wrong while the reason (R) is correct. The main components of L.P.G. are propane and butane. Methane, present in biogas can be used directly as a fuel. LPG is the abbreviation of Liquefied Petroleum Gas. Like all fossil fuels, it is a non-renewable source of energy. It is extracted from crude oil and natural gas. The main composition of LPG are hydrocarbons containing three or four carbon atoms.

34. The Pressure gauge is not given in the domestic LPG cylinders because-

- (a) They are so costly
(b) They cannot represent the quantity of gas in LPG cylinders
(c) Their use is not safe
(d) They are choked by LPGs

42nd B.P.S.C. (Pre) 1997

Ans. (b)

Oxygen and other gases found in the gas cylinders are in a gaseous state while in domestic LPG cylinders, gas is found in the liquid state so pressure gauge cannot represent the quantity of gas in LPG cylinders. That is why the pressure gauge is not given in the domestic LPG cylinders.

35. In what form cooking gas is supplied in cylinders ?

- (a) Liquid (b) Gas
(c) Solid (d) Solution

Uttarakhand Lower Sub. (Pre) 2010

Ans. (a)

See the explanation of above question.

36. Which among the following compounds is the main constituent of natural gas?

- (a) Hexane (b) Methane
(c) Butane (d) Benzene

U.P. P.C.S. (Pre) 2022

Ans. (b)

Natural gas is an odorless, gaseous mixture of hydrocarbons – predominantly made up of methane (CH_4) in addition to various smaller amounts of other higher alkanes. Low levels of trace gases like carbon dioxide, nitrogen, hydrogen sulphide, and helium are also usually present. Two forms of natural gas are currently used in vehicles and for cooking : compressed natural gas (CNG) and liquefied natural gas (LNG). Natural gas, extracted from below Earth's surface, is processed and converted to CNG and LNG.

37. Which is main component of natural gas?

- (a) Methane
- (b) Ethane
- (c) Propane
- (d) Benzene
- (e) None of these

Chhattisgarh P.C.S. (Pre) 2016

Ans. (a)

See the explanation of above question.

38. CNG is –

- (a) Compressed Natural Gas
- (b) Cyanogen Natural Gas
- (c) Condensed Nitrogen Gas
- (d) Controlled Natural Gas

U.P.P.C.S. (Pre) 2015

Ans. (a)

C.N.G. is the short form of Compressed Natural Gas. It is a readily available alternative to gasoline which is made by compressing natural gas to less than 1% of its volume at standard atmospheric pressure.

39. CNG used in automobiles to check pollution mainly consists of –

- (a) CH₄
- (b) CO₂
- (c) N₂
- (d) H₂

U.P.P.C.S. (Pre) 2005

Ans. (a)

CNG is made by compressed natural gas which is mainly composed of methane (CH₄). It is a mixture of hydrocarbons which is found deep inside Earth and have approx. 80% to 90% of methane.

40. The major component of CNG is :

- (a) CO₂
- (b) N₂
- (c) H₂
- (d) CH₄
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

See the explanation of above question.

41. Compressed Natural Gas (CNG) is mainly composed of :

- (a) Propane
- (b) Butane
- (c) Methane
- (d) Ethane

R.A.S./ R.T.S. (Pre) 2021

U.P.P.C.S. (Pre) 2022

Ans. (c)

See the explanation of above question.

42. Consider the following statements :

1. Liquefied Natural Gas (LNG) is liquefied under extremely cold temperatures and high pressure to facilitate storage or transportation in specially designed vessels.
2. First LNG terminal in India was built in Hassan.

3. Natural Gas Matters (NGM) are separated from LPG and these include ethane, propane, butane and natural gasoline.

Which of the statements given above is/are correct ?

- (a) 1 only
- (b) 1 and 3
- (c) 2 and 3
- (d) 1, 2 and 3

I.A.S. (Pre) 2005

Ans. (a)

Statement 1 is correct while statement 2 and 3 are false. Liquefied Natural Gas (LNG) is a natural gas (predominantly methane, CH₄) that has been converted to liquid form to ease of storage or transport. The natural gas is condensed into a liquid at close to atmospheric pressure by cooling it to approx. (–) 162°C.

The first LNG terminal in India was established at Dahej, Gujarat.

The main components of liquefied natural gas are generally around : methane – 80%, ethane – 7%, propane – 6%, normal butane – 4% and isobutane – 3%.

43. Oxygen is absent in :

- (a) Kerosene
- (b) Glass
- (c) Soil
- (d) Cement
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

65th B.P.S.C. (Pre) 2019

Ans. (a)

Kerosene is a combustible hydrocarbon liquid which is derived from petroleum. It is a mixture of hydrocarbons and does not contain oxygen. Oxygen is present in glass, soil and cement.

44. Gasohol is :

- (a) Ethyl alcohol + Petrol
- (b) Natural gas + Ethyl alcohol
- (c) Any gas dissolved in alcohol
- (d) Ethyl alcohol + Kerosene oil

Uttarakhand P.C.S. (Pre) 2005

Ans. (a)

Gasohol, a gasoline extender made from a mixture of gasoline (petrol) and ethanol (often obtained by fermenting agricultural crops). Gasohol has higher octane or anti-knock properties by which it burns more slowly, coolly and completely.

45. Gasohol is a mixture of :

- (a) Gasoline and Methanol
- (b) Gasoline and Ethanol
- (c) Gasoline and Propanol
- (d) Methanol and Ethanol

U.P. P.C.S. (Mains) 2016

Ans. (b)

See the explanation of above question.

46. Consider the following statements :

1. Coal, petroleum and natural gas are fossil fuels.
2. Gasohol is a mixture of benzene and alcohol.
3. Geothermal energy is non-conventional energy source.
4. Gobar gas contains mainly methane.

Of these statements :

- (a) Only 1 and 2 are correct
- (b) Only 2 and 3 are correct
- (c) Only 1, 2 and 3 are correct
- (d) 1, 3 and 4 are correct

U.P.P.C.S. (Mains) 2009

Ans. (d)

Fossil fuels are hydrocarbons primarily coal, fuel oil or natural gas formed from the remains of dead plants and animals. Sometimes, they are known as mineral fuels. Fossil fuels are fuels formed by a natural process such as anaerobic decomposition of buried dead organisms. Ethanol fuel mixtures have "E" number which describes the percentage of ethanol fuel in the mixture by volume.

E 85 is 85% denatured ethanol fuel and 15% gasoline. Low-ethanol blends from E5 to E 25 are also known as gasohol.

47. To avoid 'knocking' of the engine of a car, which one of the following is used as an anti-knocking agent?

- (a) Ethyl alcohol
- (b) Butane
- (c) Tetraethyl Lead
- (d) White Petrol

U.P.P.C.S. (Pre) 1998

Ans. (c)

Tetraethyl Lead (CH_3CH_2)₄Pb is the chief anti-knock agent for automotive gasoline or petrol. In the hot cylinder of a gasoline engine, the bonds between the lead atom and the ethyl groups are broken. Upon combustion, the lead atom forms lead oxide (PbO) which prevents fractions of the fuel mixture from burning too quickly and causing a highly undesirable engine knock. Thus, it is mixed with petrol to increase its anti-knocking rate.

48. Tetraethyl Lead (TEL) is added to petrol :

- (a) To prevent it from freezing
- (b) To increase its flash point
- (c) To increase its anti-knocking rating
- (d) To increase its boiling point

U.P.P.C.S. (Pre) 1996

Ans. (c)

See the explanation of above question.

49. Which one of the following is used as an anti-freeze for the automobile engines ?

- (a) Propyl alcohol
- (b) Ethanol
- (c) Methanol
- (d) Ethylene glycol

I.A.S. (Pre) 1997

Ans. (d)

An antifreeze is an additive, which lowers the freezing point of a water-based liquid. An antifreeze mixture is used to achieve freezing point depression for cold environments and also achieves boiling point elevation to allow higher coolant temperature. Ethylene glycol $[(\text{CH}_2\text{OH})_2]$ is used for antifreeze formulations for the automobile engines.

50. Diesel used in heavy vehicles for –

- (a) Extra mileage and save engine
- (b) Low expense and save the fuel
- (c) High power and economic saving
- (d) Being cheaper than low petrol

U.P.P.C.S. (Pre) 1992

Ans. (c)

An internal combustion engine transforms the chemical energy in fuel to mechanical rotational energy. Diesel engine, which are in existence for over a century, are the most versatile and economical engine used in a heavy vehicle. Today's direct-injection diesel engines are more rugged, powerful, durable and reliable than gasoline engines and use fuel much more efficiently as well.

51. In which of the following types of heat engines carburetor is used?

- (a) Steam engine
- (b) Diesel engine
- (c) Both diesel and petrol engines
- (d) Petrol engine

U.P. P.C.S. (Pre) 2022

Ans. (d)

A carburetor is a device used by an internal combustion engine to control and mix air and fuel entering the engine. While the earlier versions of carburetors used to do this merely by allowing air to pass over the fuel surface, the advanced version of carburetors dispense a metered amount of fuel into the stream of air to achieve this purpose. Since the 1990s, carburetors have been largely replaced by fuel injection for cars and trucks, however carburetors are still used by some small engines (e.g. lawnmowers, generators and concrete mixers) and motorcycles. Carburetors are only used in petrol engines. Diesel engines have always used fuel injection instead of carburetors.

52. Which of the following metals are used as catalyst in the Catalytic converters to eliminate poisonous gases?

- (a) Sodium
- (b) Thorium
- (c) Rhodium
- (d) None of the above

Chhattisgarh P.C.S. (Pre) 2023

Ans. (c)

